

SE 311 – Pluggable Authentication Mechanism

When designing the authentication system, our primary goal was to create a solution that is modular and extensible. To achieve this, we carefully chose suitable design patterns. Below is our thought process for pattern selection and how we mapped the participants.

1. Template Method Pattern

The authentication process for different mechanisms follows similar steps: validate credentials, retrieve user ID, and set user ID, but the implementation of these steps varies. The Template Method pattern allows us to define the skeleton of the authentication mechanism in an abstract class, while subclasses provide specific implementations.

Participants:

- Abstract Class: AuthenticationMechanism defines the template method startAuthentication() and declares abstract methods for the steps.
- Concrete Subclasses: LocalAuth, LdapAuth, and KerberosAuth implement the specific logic for each mechanism.

2. Singleton Pattern

Each authentication mechanism should have only one instance. The Singleton pattern ensures this and provides a global point of access.

Participants:

- Singleton Classes: LocalAuth, LdapAuth, and KerberosAuth each implement Singleton with a private static instance and a public static getInstance() method.

3. Iterator Pattern

The system needs to try multiple authentication mechanisms in sequence until one succeeds or all fail. The Iterator pattern provides a clean way to traverse authentication mechanisms without exposing the underlying representation.

Participants:

- We have not implemented the Iterator Pattern

4. Observer Pattern

It is important to notify other components about authentication events (successful or failed). The Observer pattern allows observers to be notified of such events.

Participants:

- We have not implemented the Iterator Pattern

5. Facade Pattern

The authentication process involves multiple steps and interactions between different components. To provide a simple interface, we used the Facade pattern. This pattern hides the complexity of the subsystem.

Participants:

- Facade Class: Authentication coordinates user input, authentication on each authentication mechanism, providing a single entry point for the authentication process. These happen in LogIn() method.

Screenshots of the Program in Execution

```
> Task :OperatingSystem.main()
Welcome, You will be logging in to your OS.
    To get started you need to enter your login credentials.
Enter your name please, then press enter:
local selin
Enter your password please, then press enter:
local doge
Authentication started for Lightweight Directory Access Protocol (LDAP) ...
Authentication started for third party authentication mechanism (Kerberos)
Authentication started on Local File System
Authentication successful in Local File System.

BUILD SUCCESSFUL in 9s
2 actionable tasks: 2 executed
5:30:56 PM: Execution finished ':OperatingSystem.main()'.

```

```
> Task :OperatingSystem.main()
Welcome, You will be logging in to your OS.
    To get started you need to enter your login credentials.
Enter your name please, then press enter:
sadasgdsa
Enter your password please, then press enter:
asdgada
Authentication started for Lightweight Directory Access Protocol (LDAP) ...
Authentication started for third party authentication mechanism (Kerberos)
Authentication started on Local File System
Your authentication failed for all authentication mechanisms.
Your login credentials are not found in any authentication mechanism.
    Therefore you are not logged in. Sorry...

BUILD SUCCESSFUL in 7s
2 actionable tasks: 1 executed, 1 up-to-date
5:34:27 PM: Execution finished ':OperatingSystem.main()'.

```

```
> Task :OperatingSystem.main()
Welcome, You will be logging in to your OS.
    To get started you need to enter your login credentials.
Enter your name please, then press enter:
ldap selin
Enter your password please, then press enter:
ldap doge
Authentication started for Lightweight Directory Access Protocol (LDAP) ...
Authentication successful in LDAP.

BUILD SUCCESSFUL in 24s
2 actionable tasks: 1 executed, 1 up-to-date
5:35:14 PM: Execution finished ':OperatingSystem.main()'.

```

> Task :classes UP-TO-DATE

```
> Task :0peratingSystem.main()
```

Welcome, You will be logging in to your OS.

To get started you need to enter your login credentials.

Enter your name please, then press enter:

kerberos selin

Enter your password please, then press enter:

kerberos doge

Authentication started for Lightweight Directory Access Protocol (LDAP) ...

Authentication started for third party authentication mechanism (Kerberos)

Authentication successful in Kerberos.

BUILD SUCCESSFUL in 11s

2 actionable tasks: 1 executed, 1 up-to-date

5:41:58 PM: Execution finished ':0peratingSystem.main()'.

UML Class Diagram

